Teacher:	Ray Joshua Balingkit	**Subject:**	Computer Science
Week:	10 (Term 1)	**Year:**	10-11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will develop their understanding of count-controlled repetition and use for loops to replace repeated instructions. They will explain the values produced by range() and use a loop to generate a multiplication table. Students will then trace an accumulator, recognise the importance of initialisation and use test evidence to identify and correct an off-by-one error.

End-of-Week Assessment/Check: How will you know they got it?

Teacher questioning, discussion of worked examples, the completed student tasks and the final checks in both lessons. Check exactly ten results and explain the start and stop values. Check the final total and explain the change needed to include the endpoint. Check that students can explain their reasoning and apply the skill without relying on the worked example.

LESSON SLOTS - MONDAY & FRIDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	26 October 2026	By the end of this lesson, students will be able to use a for loop with range to repeat a task a known number of times.	**H:** Compare five repeated print statements with one loop. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Explain count-controlled repetition and the excluded stop value in range(). Explain when the number of repetitions is known before a loop starts. Connect the values produced by range() to the number of loop iterations and show that the stop value itself is excluded. **WD:** Predict and run a loop that displays numbers 1 to 5. Use range(1, 6) and record the loop variable before each displayed result. Change the stop value and ask students to predict the number of lines before running the revised version.	**Support:** Provide the loop header and an indented print statement with one missing expression. Use a table of loop-variable values and products. Students explain why range(1, 11) is used before removing the scaffold. **Extension:** Students add a different start or step value and predict the resulting facts. They explain how the number of iterations changes and distinguish the loop variable from the chosen multiplier.

			YD: Create a multiplication-table program for a chosen number using a loop from 1 to 10. Students use one loop to generate ten labelled multiplication facts. They first predict the results for a chosen number and then check the first, last and total number of displayed lines. **C:** Check exactly ten results and explain the start and stop values. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
2	30 October 2026	By the end of this lesson, students will be able to trace a for loop and test for off-by-one errors.	**H:** Show a loop that produces only nine results. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Explain iteration, loop-variable updates and accumulating a total. Explain that an accumulator needs an initial value before the loop and is updated during each iteration. An off-by-one error often comes from misunderstanding the inclusive start or excluded stop value. **WD:** Trace a running total for the integers 1 to 4. Trace total = 0 followed by a loop adding 1, 2, 3 and 4. Record both the loop variable and running total, then compare a faulty stop value that omits the final number. **YD:** Trace and repair a total-calculation program; test one iteration and several iterations. Students repair the	**Support:** Provide the accumulator initialisation and the first trace row. Use counters to represent each added value and highlight the difference between the current loop variable and the accumulated total. **Extension:** Students investigate what happens if the total is reset inside the loop and explain the incorrect result. They justify the placement of initialisation using their trace evidence.



			endpoint and complete a trace table for the corrected program. They test a single-value interval and a longer interval, documenting the expected sum and the actual result for each. **C:** Check the final total and explain the change needed to include the endpoint. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel International GCSE Computer Science Student Book: Chapter 5
- Teacher computer and classroom display
- Classroom computers with Python 3 editor and individual save folders
- Prepared algorithm, tracing or Python practice tasks aligned with the lesson objectives
- Whiteboard and markers
- Student notebooks and pencils
- Success criteria, test tables and teacher observation checklist; confirm Friday school-event arrangements and reschedule any displaced task



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?